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Space in competitive football: A scoping review

Lisa Krampe[1], Manuel Bassek[1], Dominik Raabe[1], Daniel Memmert[1], and Robert Rein[1]

Abstract

In team sports, the ability to control key areas of the pitch and create advantageous spatial arrangements is crucial to success. However, the variety of methodological approaches to analysing space in football has resulted in inconsistent terminologies and a lack of transparency in analysis, limiting a unified understanding of space management. This scoping review aimed to examine how 'space' is conceptualised, constructed and operationalised in competitive football, resulting in a taxonomy of analytical approaches and outcomes. The review was pre-registered on the Open Science Framework (https://osf.io/6fxeb) and conducted in accordance with PRISMA-ScR guidelines. Eligible studies were identified through searches of PubMed, IEEE Xplore, and Web of Science. Inclusion criteria were English language, football-related, observational, modelling or experimental design, 11 vs. 11 adult participants, and analysis of space utilisation by individuals, groups, or teams. In total, N = 46 full-text articles on space in elite football using positional data were included. Study characteristics, space metrics, visualisation methods and modelling techniques were extracted and synthesised narratively. The review revealed conceptual and methodological heterogeneity, as similar terms were used to describe distinct constructs, identical spatial values emerged from different team formations, and the applied models showed limited validation. Further research is required to clarify the relevance and enhance the practical applicability of the spatial variables that best represent space in football and the contexts in which they are most applicable.

Keywords

Delauney triangulation, pitch zone, soccer, stretch index, tactical behaviour, Voronoi diagram

Introduction

Football is a team sport belonging to the group of invasion games, involving cooperative and competitive interactions among players at both the individual and team level.[1,2] In invasion games, how players occupy available space on the pitch is central to the successful game performance.[3,4] However, how to operationalise space and its occupation in a meaningful way is currently unclear. This has led to a large variety of approaches in the literature with overlapping terminology and interpretations, despite differing underlying constructs or alternative terms for the same construct. This scoping review aims to provide an overview and synthesis of how the concept of 'space' is conceptualised, constructed and operationalised in high performance football research.

In football, the pitch is a rectangular area and typically 105 metres long and 68 metres wide.[5] As players move, numerous behavioural options arise based on their dynamic arrangement,[6] and the players seek to create advantageous contexts, for example by overloading a pitch side to gain a numerical advantage.[7,8] Thus, players must adapt to unexpected situations and adapt their actions and tactics to the immediate context and requirements of the game.[9–11] Player movements are influenced by the team's strategy,[12] which usually include player roles, team formations,[13] and control of favorable pitch locations.[14] For example, dominating the opposition goal area can be advantageous in terms of creating goal scoring opportunities.[15,16] However, which locations are more advantageous changes throughout the game, which complicates the analysis of space in football.[6,17,18]

Of the various methodologies available in the literature, one common approach assesses teams' positions on the pitch via team length, team width, convex hull, or team spread to capture collective tactical behaviour.[18–20] For example,

Reviewer: Albin Tenga (Norwegian School of Sports Sciences, Norway)

[1]Institute of Training and Computer Science in Sport, German Sport University Cologne, Cologne, Germany

Corresponding author:
Lisa Krampe, Institute of Training and Computer Science in Sport, German Sport University Cologne, Cologne, Germany.
Email: l.krampe@dshs-koeln.de

the convex hull showed that players in ball possession were further apart and covered a larger area than the defending team.[20] Additionally, smaller convex hull areas, reflecting a more compact defensive shape,[21] were associated with fewer shots created.[17] However, these metrics do not allow the analysis of the behaviour of individual players, as aggregated values over player groupings are obtained.[22] When calculating the team length, the positions of the players closest to the own goal and opponent's goal areas used.[23] Thus, information about the players position on the pitch is disregarded, resulting in many possible arrangements of players on the pitch. Nevertheless, during attack, better teams typically exhibit greater length and width, greater stretch indices, and a larger convex hull, while, during defence, reduced width and length along with a more compact team spread and smaller convex hull are observed.[17,19,24] Although metrics like length, width and spread provide information about team dynamics and success, they hide individual player behaviour by aggregating data for entire teams. Another approach to assess space in football is based on the notion of controlling space.[25] Contrary to team-based approaches, space control assigns individual players a specific area of the pitch according to a specific segmentation model.[25,26] For example, a Voronoi diagram divides the pitch into contiguous segments based on the Euclidean distance[26] assuming that the closest player to a location can reach it before any others.[27,28] However, the model unrealistically assumes all players move and accelerate at the same velocity in every direction.[29] Accordingly, more complex models using various deterministic and probabilistic formulations have been proposed in the literature.[29,30] This includes deterministic models which assigns a single location to a player,[12] and probabilistic model which allocate a probability to each player for a specific location.[31] Player-centric approaches provide detailed insights into individual performance and strategies, while approaches that assesses probabilities for the entire team offer a holistic overview of team dynamics and tactical decisions.[32] Examples of this include the quality, creation and loss of space,[33] as well as free space on the pitch.[34] Despite the similar theoretical constructs, there is considerable variation in the methodological approaches used in these studies. Furthermore, there is still no standardised framework for assessing the validity of these approaches. The characterisation of space in football is a key focus in match analysis, yet the inconsistent use of terminology and methodological approaches hinders comparability and the integration of findings. Therefore, a scoping review of the literature was performed to provide a comprehensive and structured summary of how space is conceptualised and analysed in football research at present and to structure and synthesise the various methodological approaches and their main findings.

Methods

The scoping review followed Preferred Reporting Items for Systematic reviews and Meta- Analyses extension for Scoping Reviews (PRISMA-ScR) checklist,[35] as well as the recommended best practice guidelines for scoping reviews.[36] The protocol for this review was pre-registered on the Open Science Framework (osf.io/6fxeb).

Search strategy

Relevant literature on terms related to space in football were identified by systematically searching the electronic databases PubMed, IEEE Xplore and Web of Science up to August 2025. The search was developed and refined according to relevant synonyms, truncations and Boolean operators, which were added and optimised in accordance with the scoping review strategy.[37] This included combining the keywords ('football' or 'soccer') and ('space' or 'spatial' or 'region' or 'zone' or 'area' or 'Voronoi' or 'Delaunay' or 'triangulation'). Articles were screened for plausibility and were exported into the reference manager software Citavi (Version 6.18.0.1). After removal of duplicates, the title and abstract of the remaining studies were screened for eligibility using Rayyan.[38] Each screening followed a two-overlapping-thirds approach, ensuring that each article was independently assessed by two reviewers. Discrepancies were discussed and resolved collectively by all reviewers. All relevant articles regarding title and abstract were integrated into a full text screening. Additionally, reference lists of identified articles in the associated research field were hand-searched to detect relevant publications that were not identified by the databases.

Study selection

In order to be included in the scoping review, studies were required to fulfil the following criteria: (1) association with football, (2) English manuscripts, (3) type of study: observational, modelling and experimental, (4) 11 vs. 11 format, (5) adults, (6) types of outcome measures: space used by individuals, groups or teams. The following criteria were utilised in the exclusion of manuscripts: (1) no mention of football, (2) small-sided games, (3) focus on youth teams, (4) analysis of event-data only, (5) lack of original/primary work, (6) sole emphasis on visualisation, (7) only addressing eSports/simulation, or (8) exclusive focus on static pitch division without analysis of spatial behaviour.

Assessment of methodological quality

The methodological quality of the studies included in this scoping review was evaluated using the quality assessment tool by Letts et al.[39] Each article was independently assessed for methodological quality by at least two of the three randomly assigned reviewers using the 16-item tool. The evaluation process involved the following aspects: (1) study purpose, (2) relevant background literature, (3) appropriate design, (4) detailed sample description, (5) justified sample

Table 1. Study quality of each publication was assessed using a 16-item checklist.[39]

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	Quality score (%)
Allegre & Vuillemot[30]	1	0	1	0	0	0	0	0	1	0	0	1	0	1	1	1	44
Augusto et al.[40]	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	94
Brefeld et al.[29]	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	1	38
Bueno et al.[41]	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	88
Caetano et al.[42]	1	1	1	1	0	1	1	1	1	0	1	1	0	1	0	1	75
Caldeira et al.[43]	1	1	1	0	0	0	0	0	0	0	0	1	0	0	1	0	31
Caldeira et al.[16]	1	1	1	1	0	0	1	1	1	1	1	1	0	1	1	1	81
Carrilho et al.[12]	1	1	0	0	0	1	1	1	1	1	0	0	0	0	0	0	44
Castellano et al.[19]	1	1	0	0	1	1	1	1	0	0	0	1	0	1	0	0	50
Cerda et al.[15]	1	1	1	0	0	0	1	0	0	0	0	1	1	0	1	1	50
Clemente et al.[44]	1	1	1	0	0	0	0	1	1	1	1	0	0	1	0	1	56
Clemente et al.[45]	1	1	1	0	0	0	0	1	1	1	1	0	0	1	0	1	56
Duarte et al.[46]	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1	1	88
Errekagorri et al.[47]	1	1	1	1	0	1	1	1	1	0	0	1	0	1	1	1	75
Forcher et al.[48]	1	1	1	1	0	1	0	1	1	1	1	1	0	1	1	1	81
Forcher et al.[49]	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1	1	88
Forcher et al.[24]	1	1	1	0	0	1	0	0	1	0	0	1	0	0	1	1	50
Garcia-Calvo et al.[18]	1	1	1	0	0	1	0	0	1	1	1	1	1	1	1	1	75
Goncalves et al.[22]	1	1	0	0	0	1	0	0	1	0	0	1	0	0	0	0	31
Goncalves et al.[50]	1	1	1	0	0	1	0	0	1	0	1	1	1	1	1	1	69
Gu et al.[51]	1	1	0	0	0	0	1	0	1	0	1	1	0	0	0	0	38
Guan et al.[21]	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	88
Knudsen et al.[52]	1	0	0	0	1	0	0	0	1	1	1	1	1	1	1	1	63
Knudsen et al. [53]	1	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	25
Knudsen et al.[54]	1	1	0	0	0	1	0	0	1	0	0	1	1	1	1	0	50
Külah et al.[55]	1	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	38
Lorenzo-Martinez et al.[56]	1	0	1	0	0	0	0	0	1	1	1	0	0	1	0	0	38
Martens et al.[8]	1	1	1	1	0	0	1	1	1	1	1	1	0	1	1	0	75
Memmert et al.[57]	1	1	1	1	0	1	1	0	1	1	1	1	0	1	1	1	81
Memmert et al.[58]	1	1	0	1	0	1	1	0	1	0	1	1	0	1	0	1	63
Merlin et al.[59]	1	1	1	1	0	1	0	0	1	1	1	1	1	1	1	1	81
Moura et al.[20]	1	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	25
Moura et al.[60]	1	1	1	0	0	1	0	0	1	1	1	1	0	1	1	0	63
Moura et al.[61]	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	1	88
Narizuka et al.[6]	1	1	1	0	0	1	0	0	1	1	1	1	0	1	0	0	56
Narizuka et al.[62]	1	1	1	0	0	0	0	0	1	1	1	0	0	1	0	0	44
Oetting et al.[63]	1	1	1	1	0	0	1	1	1	1	1	1	0	1	1	1	81
Passos et al.[64]	1	1	1	0	0	0	0	0	1	1	1	0	0	1	0	0	44
Praca et al.[17]	1	1	1	1	0	0	1	1	1	1	1	1	0	1	1	1	81
Rein et al.[65]	1	1	1	0	0	0	0	1	1	1	1	1	0	1	1	1	69
Spearman et al.[31]	1	1	1	0	0	0	0	1	1	1	1	1	0	1	1	1	69
Spearman et al.[66]	1	0	1	0	0	0	0	0	1	0	0	1	0	0	1	1	38
Stein et al.[34]	1	0	1	0	0	0	0	0	1	0	0	1	0	0	1	1	38
Taki et al.[25]	1	1	1	0	0	0	1	1	1	0	1	1	0	1	1	0	63
Taki et al.[67]	1	1	0	1	0	0	0	0	0	0	0	1	0	0	1	1	38
Zubillaga et al.[68]	1	1	1	0	0	1	0	0	1	1	1	1	1	1	1	1	75

size, (6) informed consent obtained, (7) reliable outcome measures, (8) valid outcome measures, (9) detailed method description, (10) appropriate statistical analysis, (11) appropriate analysis methods, (12) significance for practice, (13) reported drop-outs, (14) appropriate conclusions, (15) practical actionable measures, (16) acknowledged limitations. These criteria were scored on a binary scale 1 = yes (was included in the article), 0 = no (was not included in the article or cannot determine, not applicable, not reported). A quality score was then calculated for each study by summing its binary scores and dividing the result by the maximum attainable score. This was expressed as a percentage in order to reflect a measure of methodological quality. (1) Low methodological quality for scores ≤25%; (2) good methodological quality for

Table 2. Study meta-characteristics – studies listed in reverse chronological order by year, followed by alphabetical order for published studies.

Study	Type of study	Sport or computer science	Gender	Gender reported by authors	Number of matches	Professional vs amateur	Competition
Allegre & Vuillemot[30]	Technological	Computer science	Male	Yes	2 goals	Professional	English Premier League, Quarter final Champions League
Augusto et al.[40]	Empirical	Sport science	Male	No	17 matches	Professional	1st Division Brazilian National Football League
Brefeld et al.[29]	Empirical	Computer science	NA	No	1 match	Professional	German Bundesliga
Bueno et al.[41]	Technological	Computer science	Male	No	11 matches	Professional	1st Division Brazilian National Football League
Caetano et al.[42]	Empirical	Sport science	Male	Yes	4 matches	Professional	Sao Paulo State league
Caldeira et al.[43]	Observational	Sport science	Male	No	10 matches	Professional	French Ligue 1
Caldeira et al.[16]	Empirical	Sport science	Male	No	283 matches	Professional	French Ligue 1, Champions League
Carrilho et al.[12]	Empirical	Sport science	Male	No	1 match	Professional	FIFA World Cup
Castellano et al.[19]	Empirical	Sport science	Male	No	6 matches	Professional	Spanish La Liga
Cerda et al.[15]	Empirical	Sport science	Male	No	94 goals in 64 matches	Professional	FIFA World Cup 2018
Clemente et al.[44]	Technological	Sport science	NA	No	3 matches	Professional	NA
Clemente et al.[45]	Empirical	Sport science	NA	No	3 matches	Professional	NA
Duarte et al.[46]	Empirical, technological	Sport science	Male	Yes	1 match	Professional	English Premier League
Errekagorri et al.[47]	Empirical	Sport science	Female	Yes	8 matches	Semi-professional	2nd Women's Spanish Division
Forcher et al.[48]	Technological	Computer science	Male	No	153 matches	Professional	German Bundesliga
Forcher et al.[49]	Empirical, observational	Sport science	Male	No	153 matches	Professional	German Bundesliga
Forcher et al.[24]	Observational	Sport science	Male	No	153 matches	Professional	German Bundesliga
Garcia-Calvo et al.[18]	Empirical	Sport science	NA	No	760 matches	Professional	Spanish La Liga
Goncalves et al.[22]	Observational	Sport science	NA	No	12 matches	Professional	English Premier League
Goncalves et al.[50]	Empirical	Sport science	NA	No	First half of 6 matches	Professional	Preseason Portuguese Primeira Liga
Gu et al.[51]	Observational	Sport science	Male	No	43 matches	Professional	English Premier League
Guan et al.[21]	Technological	Sport science	Male	No	237 matches	Professional	Chinese Super League
Knudsen et al.[52]	Observational	Sport science	NA	No	1 match	Professional	Danish Superliga
Knudsen et al.[53]	Empirical	Sport science	Male	Yes	4 matches	Professional	Danish Superliga
Knudsen et al.[54]	Observational	Sport science	NA	No	10 matches	Professional	Danish Superliga
Külah et al.[55]	Observational	Computer science	Male	Yes	306 matches	Professional	Turkish Super League
Lorenzo-Martinez et al.[56]	Observational	Sport science	NA	No	234 matches	Professional	German Bundesliga
Martens et al.[8]	Observational	Computer science	NA	No	54 matches	Professional	German Bundesliga
Memmert et al.[57]	Empirical	Sport science	Male	Yes	144 trials	Amateur	NA
Memmert et al.[58]	Empirical	Sport science	Male	Yes	177 trials	Amateur	higher amateur level
Merlin et al.[59]	Observational	Sport science	NA	No	4 matches	Professional	1st Division Brazilian National Football League

(continued)

Table 2. Continued.

Study	Type of study	Sport or computer science	Gender	Gender reported by authors	Number of matches	Professional vs amateur	Competition
Moura et al.[20]	Empirical	Sport science	Male	No	10 matches	Professional	1st Division Brazilian National Football League
Moura et al.[60]	Empirical	Sport science	Male	No	8 matches	Professional	1st Division Brazilian National Football League
Moura et al.[61]	Empirical	Sport science	NA	No	14 matches	Professional	1st Division Brazilian National Football League
Narizuka et al.[6]	Observational	Computer science	NA	No	54 matches	Professional	Japan Football League
Narizuka et al.[62]	Observational	Sport science	NA	No	45 games	Professional	Japan Football League
Oetting et al.[63]	Technological	Computer science	NA	NA	3 matches	NA	NA
Passos et al.[64]	Empirical	sport Science	NA	No	1 match	Professional	Dutch Eredivisie
Praca et al.[17]	Empirical	Sport science	male	No	275 matches	Professional	German Bundesliga
Rein et al.[65]	Empirical	Sport science	NA	No	103 matches	Professional	German Bundesliga
Spearman et al.[31]	Technological	Computer science	Male	No	38 matches	Professional	English Premier League
Spearman et al.[66]	Technological	Computer science	NA	No	58 matches	Professional	NA
Stein et al.[34]	Technological	Sport science	NA	No	65 matches	Professional	Different 1st Leagues and Competitions
Taki et al.[25]	Technological	Computer science	NA	No	NA	NA	NA
Taki et al.[67]	Technological	Computer science	NA	No	NA	NA	NA
Zubillaga et al.[68]	Empirical	Sport science	Female	Yes	4 matches	Professional	1st Women's Spanish Division

Study	Type of tracking system	Tracking frequency (Hz)	Additional data sources	Space terminology	Independent variables	Data availability	Algorithms/ code availability
Allegre & Vuillemot[30]	NA	NA	NA	Disputed area	NA	Raw data	Yes
Augusto et al.[40]	GPS	10	NA	Team length/width, LpW, team spread, space exploration index, stretch index	Playing time, match location, ranking of opposition, match outcome, match status	NA	No
Brefeld et al.[29]	NA	25	NA	Zones of control	NA	NA	Yes
Bueno et al.[41]	Video semi-automatic	NA	NA	Convex hull	Shape description	Available	Yes
Caetano et al.[42]	Video semi-automatic	30	NA	Dominant region	Prediction accuracy	NA	Yes
Caldeira et al.[43]	Video semi-automatic	10	NA	EPS/Relative Space per Player	Team formation, ball possession status, players role	NA	No
Caldeira et al. [16]	Video semi-automatic	NA	Notational data	Finishing space value, player relative Voronoi	Distance/angle to centre of goal line, distance ball carrier to other players	NA	No

(continued)

Table 2. Continued.

Study	Type of tracking system	Tracking frequency (Hz)	Additional data sources	Space terminology	Independent variables	Data availability	Algorithms/ code availability
Carrilho et al. [12]	Video semi-automatic	25	NA	Convex hull, player-ball-goal angles	Ball possession, halves of the match	NA	Yes
Castellano et al. [19]	Video semi-automatic	NA	NA	Team length/width, surface area	Team strength, possession, ApEn	NA	Yes
Cerda et al.[15]	Video semi-automatic	NA	Events of goals scored	Pitch zones	Team possession type (counter attack vs. elaborate play)	NA	No
Clemente et al.[44]	Video semi-automatic	30	NA	Weighted centroid, weighted stretch index, surface area, effective area of play	Ball possession status, halves of the match	NA	No
Clemente et al.[45]	Video semi-automatic	30	NA	Weighted centroid, weighted stretch index, surface area, effective area of play	Score status	NA	No
Duarte et al.[46]	Video automatic	10	NA	Surface area, stretch index, team length/width, geometrical centre	Goals, game breaks	NA	No
Errekagorri et al.[47]	GPS	10	NA	Team width/length/height, surface area	Time period, distance covered	NA	No
Forcher et al.[48]	Video semi-automatic	25	NA	Surface area, team spread numerical superiority (static/dynamic pitch zones)	Inter-line distances, defensive pressure, distance to ball, player velocity	NA	Yes
Forcher et al.[49]	Video semi-automatic	25	Event data, interview	Surface area, team length, team spread, space control	Defensive pressure, distance to ball, player velocity	NA	No
Forcher et al.[24]	Video automatic	25	Event data	surface area	(Un-) successful defensive actions	NA	No
Garcia-Calvo et al.[18]	Video automatic	NA	NA	Convex hull	Pre/post lockdown; home/ away	NA	No
Goncalves et al.[22]	Video semi-automatic	25	NA	EPS, game width, game length	Team strength, number of opponents	NA	Yes
Goncalves et al.[50]	GPS	5	NA	EPS	Size of sub-group (3–10 players)	NA	Yes
Gu et al.[51]	NA	Not reported, down sampled to 1	Event data	Pitch control	Predicted vs actual possession value (possession evaluation)	NA	Partially
Guan et al.[21]	Video semi-automatic	10	Event data	Convex hull	Possession, time, goal differential, betting odds	NA	No
Knudsen et al.[52]	LPS	20	NA	Coverable area	NA	NA	Yes
Knudsen et al.[53]	LPS	20	NA	Reachable area	Prediction accuracy	NA	Yes
Knudsen et al.[54]	LPS	20	NA	Possible region	NA	NA	Yes
Külah et al.[55]	Video semi-automatic	NA	NA	Player influence area, pass interception area, team spread	Possession	NA	Yes

(continued)

Table 2. Continued.

Study	Type of tracking system	Tracking frequency (Hz)	Additional data sources	Space terminology	Independent variables	Data availability	Algorithms/ code availability
Lorenzo-Martinez et al.[56]	Video semi-automatic	25	NA	Team centroid, inter-team centroid distance, team length/width, LpW, stretch index, Voronoi, space control	Substitutions, pitch third	NA	Yes
Martens et al.[8]	Video semi-automatic	25	NA	Player influence area, pitch control	NA	NA	Yes
Memmert et al.[57]	LPS	25	NA	EPS, LpW, space control gain, convex hull	Team formation	NA	Yes
Memmert et al.[58]	LPS	25	NA	EPS, space control gain	High vs low attention capacity	NA	No
Merlin et al.[59]	Video semi-automatic	15	NA	EPS	Ball possession	NA	Yes
Moura et al.[20]	Video automatic	30	Hand-tagged ball possession	Coverage area	Halves of the match	NA	No
Moura et al.[60]	Video automatic	30	Hand tagged events of shots and tackles	Convex hull, team spread, surface area	Shots, tackles, ball possession	NA	No
Moura et al.[61]	Video semi-automatic	NA	NA	suRface area	Pass, tackle, shots to the goal, goals scored, ball possession	Available	Yes
Narizuka et al.[6]	Video semi-automatic	25	NA	Arrival region	NA	NA	Yes
Narizuka et al.[62]	Video semi-automatic	25	NA	Safe/risky sparse/dense space	NA	NA	No
Oetting et al.[63]	Video semi-automatic	25	NA	EPS	Classification of tactical formations	Available	Yes
Passos et al.[64]	Video semi-automatic	25	NA	Space-time spread	NA	NA	Yes
Praca et al.[17]	Video semi-automatic	10	NA	Team length/width, LpW, stretch index, near the ball space control	Halves of the match, goal difference, match outcome	NA	No
Rein et al.[65]	Video semi-automatic	25	NA	Space control	Passing effectiveness	NA	Yes
Spearman et al.[31]	NA	25	Event data	Pitch control	(Un-) successful passes, goals vs saved shots after corner	NA	No
Spearman et al.[66]	NA	25	Event data	Pitch control	NA	NA	No
Stein et al.[34]	Video automatic	10	Event data	Free space, interaction space	NA	NA	No
Taki et al.[25]	Video semi-automatic	NA	Pitch markings for transformation	Minimum moving time, dominant region	Position, velocity, acceleration, ball vector	NA	No
Taki et al.[67]	Video semi-automatic	NA	NA	Dominant region	NA	NA	No
Zubillaga et al.[68]	Video automatic	25	NA	(Individual-) playing area	Pitch zones	NA	No

Table 3. Space conceptualisation - overview of studies by model formulation.

Study	Space conceptualisation	Pitch-, player-, or team-based	Model description (verbal, sketch, technical)	Reproducibility	Model variables	Mathematical model
Allegre & Vuillemot[30]	Overlapping area	Pitch & player	Sketch	Yes	Player position, velocity, orientation	Probabilistic
Augusto et al.[40]	Team width/length, LpW, spread, stetch index, space exploration index	Player & team	Verbal	Yes	Player position	Deterministic
Brefeld et al.[29]	Dominant region	Player	Technical	Yes	Player position, velocity	Probabilistic
Bueno et al.[41]	Convex hull	Team	Sketch	Yes	Player position	Deterministic
Caetano et al.[42]	Dominant region	Player	Technical	Yes	Player position, velocity	Probabilistic
Caldeira et al.[43]	Space control	Player	Sketch	Yes	Player position	Voronoi
Caldeira et al.[16]	Space control, player-goal distances & angles	Player & pitch	Sketch	Yes	Player position	Deterministic
Carrilho et al.[12]	Convex hull, player-ball-goal angles	Pitch & team	Sketch	Yes	Zone position	Deterministic
Castellano et al.[19]	Surface area	Team	Technical	Yes	Player position	Deterministic
Cerda et al.[15]	Pitch zones	Pitch	Sketch	Yes	Scored from pitch zone, number of players in pitch zones	Decision tree notes
Clemente et al.[44]	Team centroid, stretch index, convex hull	Team	Sketch	Yes	Player position	Deterministic
Clemente et al.[45]	Team centroid, stretch index, convex hull	Team	Sketch	Yes	Player position	Deterministic
Duarte et al.[46]	Convex hull, stretch index, team length/ width, team centroid	Team	Sketch	Yes	Player position	Deterministic
Errekagorri et al.[47]	Team length/width/height, convex hull	Team	Sketch	Yes	Player position	Deterministic
Forcher et al.[48]	Convex hull, team spread, numerical superiority (static/dynamic pitch zones)	Team, player, pitch	Technical	Yes	Player position, velocity	Probabilistic
Forcher et al.[49]	Team length/width/height, team spread, space control, convex hull, numerical superiority	Player, team, pitch	Technical	Yes	Player position	Probabilistic
Forcher et al.[24]	Compactness of defensive organisation	Team	Verbal	Yes	Player position	Linear mixed model
Garcia-Calvo et al.[18]	Convex hull	Team	Verbal	No	Player position	Linear mixed model
Goncalves et al.[22]	Convex hull	Team	Technical	Yes	Player position	Deterministic
Goncalves et al.[50]	Convex hull	Team	Technical	Yes	Player position	Deterministic
Gu et al.[51]	Dominant region	Pitch & player	Technical	Yes	Player position, velocity, facing direction, player ID	Probabilistic
Guan et al.[21]	Convex hull	Team	Technical	Yes	Player position	Deterministic
Knudsen et al.[52]	Dominant region	Player	Technical	Yes	Player position	Deterministic
Knudsen et al.[53]	Dominant region	Player	Sketch	Yes	Player position	Deterministic
Knudsen et al.[54]	Dominant region	Player	Technical	Yes	Player position	Deterministic
Külah et al.[55]	Dominant region	Player	Technical	Yes	Player position, ball position	Deterministic & statistical

(continued)

Table 3. Continued.

Study	Space conceptualisation	Pitch-, player-, or team-based	Model description (verbal, sketch, technical)	Reproducibility	Model variables	Mathematical model
Lorenzo-Martinez et al.[56]	Space control	Player & team	Sketch	Yes	Player position	Deterministic
Martens et al.[8]	Dominant region	Pitch & player	Technical	Yes	Player position	Probabilistic
Memmert et al.[57]	Convex hull, space control gain	Player & team	Sketch	Yes	Player position	Deterministic
Memmert et al.[58]	Space control gain	Team	Verbal	Yes	Player position	Deterministic
Merlin et al.[59]	Convex hull	Team	Technical	Yes	Player position	Deterministic
Moura et al.[20]	Convex hull	Team	Sketch	No	Player position	Deterministic
Moura et al.[60]	Convex hull	Team	Sketch	No	Player position	Deterministic
Moura et al.[61]	Convex hull	Team	Technical	Yes	Player position	Statistical
Narizuka et al.[6]	Dominant region	Player	Technical	Yes	Player position	Probabilistic
Narizuka et al.[62]	Safe/risky sparse/dense space	Team	Technical	Yes	Player position	Statistical
Oetting et al.[63]	Convex hull	team	Technical	Yes	Player position	Deterministic
Passos et al.[64]	Dominant region	Pitch	Sketch	Yes	Player position, ball position	Deterministic
Praca et al.[17]	Team length/width, LpW, stretch index, near-the-ball space, numerical control of space	Team & player	Technical	Yes	Player position	Probabilistic
Rein et al.[65]	Space control	Player & team	Sketch	Yes	Player position, ball position	Deterministic
Spearman et al.[31]	Dominant region	Player	Technical	Yes	Player position, velocity, acceleration	Probabilistic
Spearman et al.[66]	Dominant region	Pitch & player	Technical	Yes	Player position, velocity, acceleration	Probabilistic
Stein et al.[34]	Overlapping area, free space	Pitch	Technical	No	Player position, velocity, heading	Probabilistic
Taki et al.[25]	Dominant region	Player, team	Sketch	No	Player position, velocity, acceleration	Deterministic
Taki et al.[67]	Dominant region	Player	Technical	Yes	Player position, velocity, acceleration	Probabilistic
Zubillaga et al.[68]	Individual surface area	Pitch & team	Sketch	Yes	Player position	Deterministic

Table 4. Space conceptualisation – overview of studies by analysis conducted.

Study	Type of analysis	Contextual variables	Space variable	Construct validity
Allegre & Vuillemot[30]	Qualitative	NA	NA	Expert assessment
Augusto et al.[40]	Quantitative	Playing time, match location, ranking of opposition, match outcome, match status	Dependent	NA
Brefeld et al.[29]	Qualitative	NA	NA	NA
Bueno et al.[41]	Quantitative	Player position	Team formation shape description	Face
Caetano et al.[42]	Quantitative	NA	Independent	Prediction
Caldeira et al.[43]	Quantitative	NA	Dependent	NA
Caldeira et al.[16]	Quantitative	Opposition context	Independent	Expert assessment
Carrilho et al.[12]	Quantitative	Ball possession, halves of the match	Independent	NA
Castellano et al.[19]	Quantitative	Team strength, possession	Both	NA
Cerda et al.[15]	Quantitative	Decision tree notes	Independent	Prediction
Clemente et al.[44]	Quantitative		Dependent	NA
Clemente et al.[45]	Quantitative	Halves of the match, ball possession	Dependent	NA
Duarte et al.[46]	Quantitative	Playing time period, team (home/away)	Dependent	NA
Errekagorri et al.[47]	Quantitative	Time period (15min)	Dependent	NA
Forcher et al.[48]	Quantitative	Sucessful (rest) defence, ball recovery	Independent	Prediction
Forcher et al.[49]	Mixed-methods (expert interview & machine learning)	Defensive success, defensive pressure, distance to ball, player speed, inter-line distances, numerical superiority	Independent	Prediction, expert assessment
Forcher et al.[24]	Quantitative	NA	Dependent	NA
Garcia-Calvo et al.[18]	Quantitative	Pre/post lockdown; home/away	Dependent	Reference to other work
Goncalves et al.[22]	Quantitative	Team strength	Dependent	Face
Goncalves et al.[50]	Quantitative	Size of sub-group	Dependent	Face
Gu et al.[51]	Quantitative	NA	Dependent	Face
Guan et al.[21]	Quantitative	Ball possession, ball (not) in play, betting odds, goal differential	Dependent	NA
Knudsen et al.[52]	Quantitative	NA	NA	Face
Knudsen et al.[53]	Quantitative	NA	Independent	Face
Knudsen et al.[54]	Quantitative	NA	Dependent	Face
Külah et al.[55]	Quantitative	NA	Independent	Face
Lorenzo-Martinez et al.[56]	Quantitative	Match location, team strength, match status	Dependent	Face
Martens et al.[8]	Quantitative	Player velocity and direction, ball position, opponent pressure	Dependent	Prediction
Memmert et al.[57]	Quantitative	Team formation	Dependent	Face
Memmert et al.[58]	Quantitative	NA	Dependent	Reference to other work
Merlin et al.[59]	Quantitative	NA	Dependent	Face
Moura et al.[20]	Quantitative	NA	Dependent	NA
Moura et al.[60]	Quantitative	Ball possession	Dependent	NA
Moura et al.[61]	Quantitative	Ball possession	Dependent	Correlation
Narizuka et al.[6]	Quantitative	NA	Dependent	Prediction
Narizuka et al.[62]	Quantitative	NA	Dependent	NA
Oetting et al.[63]	Quantitative	Scoreline, time series, ball possession	Dependent	NA
Passos et al.[64]	Quantitative	NA	Independent	Expert assessment
Praca et al.[17]	Quantitative	Match outcome, goal difference, halves of the match	Dependent	NA
Rein et al.[65]	Quantitative	Match location, team strength	Independent	Face

(continued)

Table 4. Continued.

Study	Type of analysis	Contextual variables	Space variable	Construct validity
Spearman et al.[31]	Quantitative	NA	Both	Prediction
Spearman et al.[66]	Quantitative	NA	Dependent	Prediction
Stein et al.[34]	Qualitative	NA	NA	Expert assessment
Taki et al.[25]	Quantitative	NA	Dependent	Face
Taki et al.[67]	Qualitative	NA	NA	Face
Zubillaga et al.[68]	Quantitative	NA	Dependent	NA

scores between 51% and 75%; and (3) excellent methodological quality for scores >75%. The scoring and classification methods employed here are consistent with those used in other reviews (Table 1).[2]

Data extraction

Two reviewers extracted the relevant data from each study independently. Any discrepancies between the two reviewers were first discussed and, if no resolution was reached, a third reviewer was consulted to verify inconsistencies. The studies were categorised based on various meta-characteristics (Table 2). These include the author(s), publication year, study type, sport or computer science, gender, gender reported by authors, number of matches, professional vs. amateur, competition, type of tracking system, tracking frequency (Hz), additional data sources, space terminology, independent variables, data availability and algorithms/code availability. The conceptualisation of space is addressed through two levels of analysis: the analytical framework (Table 3) and the model formulation (Table 4). Table 3 outlines space conceptualisation, including pitch-, player-, team-based approach, model description and reproducibility, model variables and mathematic formulation. Table 4 presents the main findings, detailing the type of analysis conducted, contextual and spatial variables considered and the construct validity of the models.

Results

Research results

The initial research yielded a total of 8.863 studies, of which 2.808 were identified as duplicates. Following the removal of duplicates, 6.055 studies were subjected to the initial screening process. The initial screening process, based on the title and abstract, resulted in the exclusion of 5.965 studies, with 90 studies proceeding to the full-text screening stage. The primary reasons for exclusion were: no football ($n = 1$), no 11 vs. 11 ($n = 1$), small-sided games ($n = 6$), only youth ($n = 1$), only event data ($n = 7$), not original/primary work ($n = 12$), only visualisation ($n = 6$), only eSports/simulation ($n = 1$) and no assignment to pitch/player/team ($n = 10$). Furthermore, one study was selected manually following the screening process, resulting in a total of 46 studies being included in the analysis. The flow chart systematically illustrates the literature search process (Figure 1).

Methodological quality of the studies

In the evaluation of methodological quality, the mean quality score of the included studies was 58%. Scores ranged from a minimum of 25% to a maximum of 94%. A total of 19 studies were classified with a low methodological quality (quality score <50%), while 15 studies were rated as good (quality score between 51% and 75%) and 12 studies were rated as excellent methodological quality (quality score >75%) (Table 1).

Study characteristics of included studies

The included studies reported data from various leagues, including the English Premier League (EPL) ($n = 6$), German Bundesliga ($n = 8$), Brazilian Serie A2 Sao Paulo State League ($n = 7$), French Ligue 1 ($n = 2$), Netherland Eredivisie ($n = 1$), FIFA World Cup 2018 ($n = 2$), Spanish La Liga ($n = 2$), Portuguese Primeira Liga ($n = 1$), Danish Superliga ($n = 3$), Turkish Super Lig ($n = 1$), Japanese J1 League ($n = 2$), Chinese Super League ($n = 1$), Champions League ($n = 1$) and even from the 1st and 2nd Womens's Spanish Division ($n = 2$). Additionally, some studies included data from amateur teams ($n = 2$) or they did not mention any further information in relation to the exact team ($n = 8$). It is notable that in 37 of the 46 (80%) studies, no information regarding gender was provided. Data collection periods varied, ranging from congested periods spanning a single game to an entire seasons. The analysis included articles published from 1996 to 2024.

Discussion

The objectives of the review were to examine how space has been conceptualised and analysed in football, how their methodological procedures have been constructed and what their predictive accuracy is. To answer these questions, this review aimed to provide a comprehensive overview and

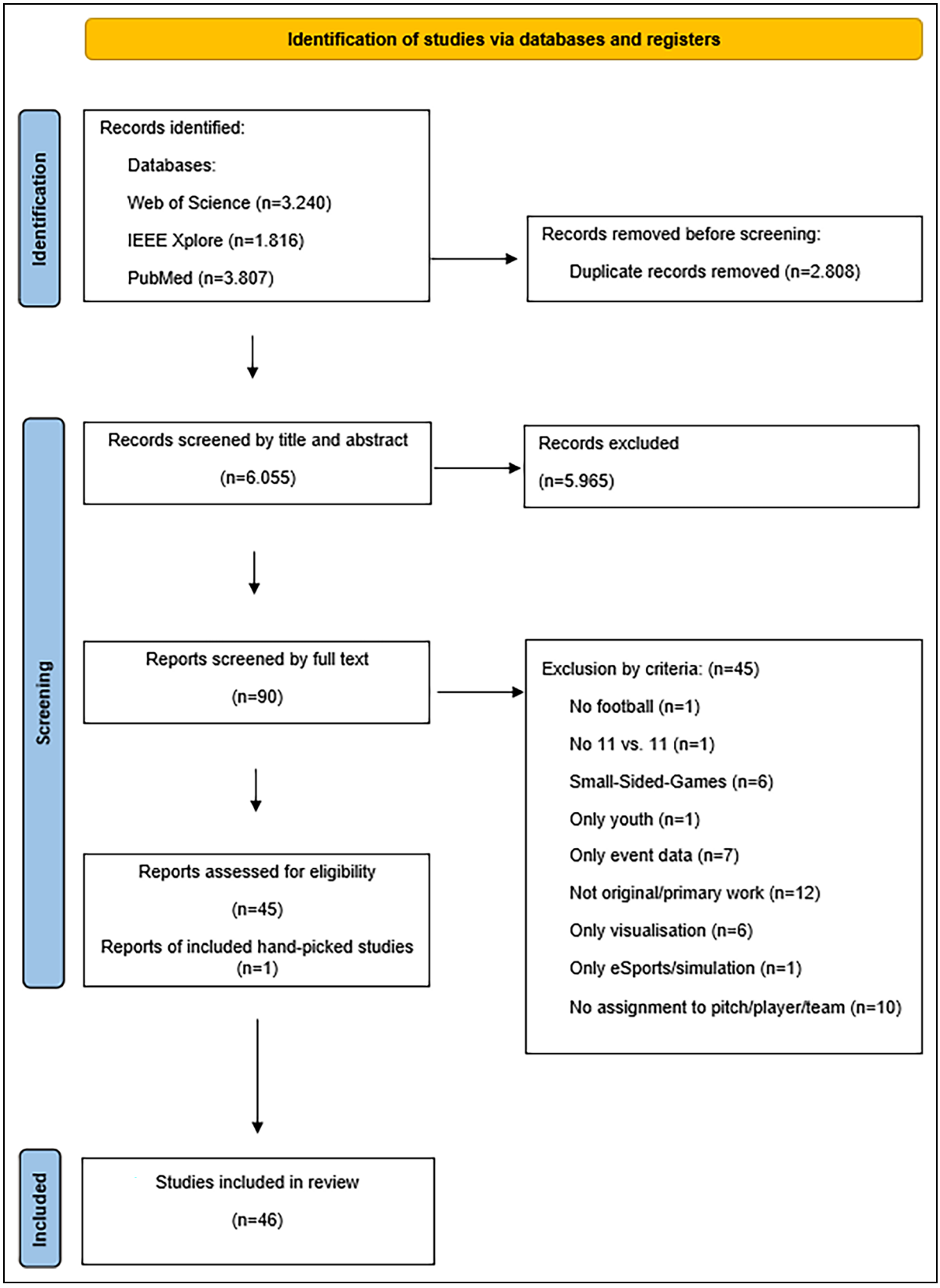

Figure 1. Flow chart of study selection process.

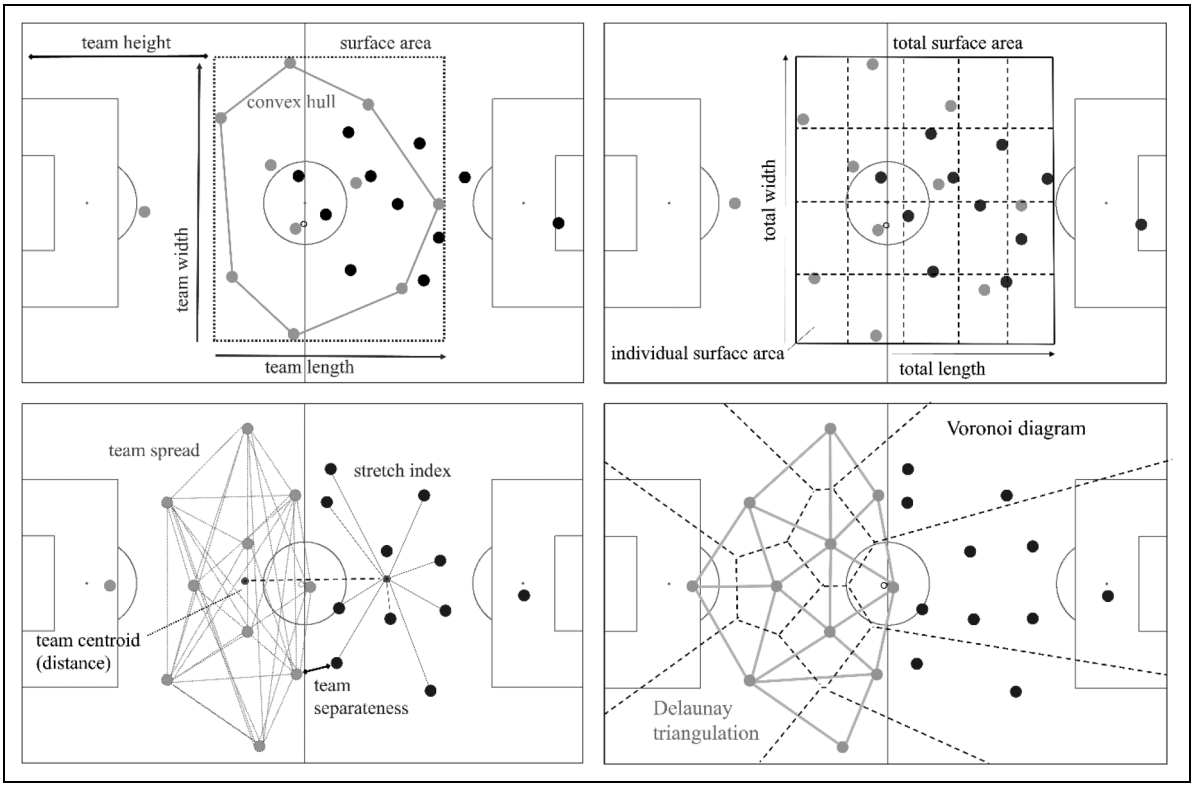

Figure 2. Metrics of collective tactical behaviour of players in space are displayed. These include measures such as team length, team width, team height, team surface area, convex hull, team centroid, inter-team centroid distance, team spread, stretch index, team separateness, Voronoi diagram and Delaunay triangulation.

synthesis of how the concept of 'space' is constructed and operationalised in high-performance football research. The narrative focus of this review was on the organisation of methods and the results, based on continuous spatiotemporal data.

Collective tactical behaviour

Simplified metrics of spatial organisation are the *team's length [m]* and *width [m]* ($n = 5$ studies), *team height [m]* ($n = 2$ studies) and the *length-to-width ratio (LpW)* ($n = 2$ studies). Team length is the distance between the two players furthest apart on the long axis of the pitch; team width is the distance across the short axis.[23] The team height is the mean team defence depth per match, understood as the distance between the furthest back player and the goal being defended.[47] The LpW is defined as the ratio of the team length to the team width (Figure 2).[1]

Evidence suggests that offensive formations during long ball possessions show greater offensive team width than shorter possessions.[59] Defensive length and width were higher against stronger teams (top six), while offensive length and width increased against weaker teams (bottom seven).[19,40] Successful teams showed greater defensive length during high pressing in the first half, reflecting field expansion and greater attacking width.[17] While team length became more predictable over time, width was affected by goals and breaks.[46] However, no significant changes in width, length or height were found across different match phases.[47] Interestingly, teams with a greater attentional capacity had a narrower width but similar length, producing greater LpW.[58] Team length increased after offensive substitutions, while LpW increases with neutral and offensive changes,[56] and during attacking play.[17] A 3-5-2 formation yielded a greater LpW than the 4-2-3-1 formation,[57] and greater LpW was observed in the second half and when teams were losing.[40] In summary, team length and width indicate how far players are positioned vertically and horizontally. The LpW reflects the team's shape and spatial balance. Offensively, greater length and width creates space, stretches defences and enables penetrating passes.[69] However, as these metrics consider only the two most distant outfield players, a single player may distort the team structure, and different formations can yield identical length or width values.

A more refined geometric approach to space is the team's *convex hull [m^2] ($n = 18$)*.[70] The convex hull is defined as the smallest convex polygon area bounded by the edges of the outfield players.[22] In some studies ($n = 8$) the team's convex hull is termed as the *surface area*. Castellano et al.[19]

introduced the surface area of a team, as the rectangle formed by the team's length and width around all team's furthest outfield players. In addition, the convex hull is also referred as *effective playing space (EPS*) given in absolute EPS (m^2) ($n = 7$ studies)[57,59] and as a relative EPS (%) ($n = 2$ studies)[22,50] of the pitch. The *effective area of play* is defined as the space where the convex hulls of both teams overlap ($n = 2$ studies).[44] It is also notable, that the term *surface area*, is applied to different constructs, referring both to the *convex hull* and to the *product of team's length and width.* To avoid inconsistencies, the term *convex hull* (m^2) is used here for clarity and consistency with previous studies. The calculation of the *surface area* as length multiplied by width remains described under this term.

The convex hull is influenced by contextual factors, like match progression and score line.[44,45] Changes to the convex hull appear to reflect key events and temporal dynamics.[46] Smaller convex hulls were observed in less successful teams,[50] whereas during the COVID pandemic lockdown average convex hulls were reduced across teams.[18] Attacking teams occupy greater surface areas (length × width) against weaker opponents (bottom six).[19] No significant differences were found when comparing two team formations.[57] The overlap between the two teams' convex hulls was smaller in the second half than in the first, and this was influenced by score status.[44,45] Using Multiscale Fractal Dimension, Moura et al.[61] suggest that the complexity of convex hull of the outfield players reveals recurrent formation patterns linked to tactical technical performance.[41] Hidden Markov modelling further indicates collective behaviours of both teams, identifying latent states of attacking, defending and pressing that corresponded to formation variation and match events.[63] Gonçalves et al.[50] extend the construct of team length and team width by defining the *game width* and *game length* as the length and width of all outfield players on both teams. *Game space* is defined as the product of game length and game width[50] (Figure 2). Zubillaga et al.[68] used the term *playing area* for the same calculation used for *game space.*[68] In the following, the previously described calculation will be referred to as the *total surface area,* which is made up of the *total length* and *total width* of all outfield players. The total surface area increased when playing against lower-ranked opponents (bottom three), likely as a tactical approach to exploiting lateral space on the pitch in order to disrupt the defensive stability of these teams.[50] An *individual playing area* was calculated by dividing the total surface area of the pitch equally among the players, allocating each an identical portion of the overall space.[68] However, this approach simplifies the complexity of football tactics, neglecting positional roles, movement patterns, and boundary lines, making it an unrealistic representation of actual game dynamics. Depending on the position of the ball, the individual surface area was significantly smaller in the midfield positions and larger near the goal than in the other defined pitch zones.[68] In summary, the team`s surface area is frequently overestimated in comparison to the convex hull, as its calculation incorporates solely the two most distant players. While the total surface area of both teams does represent the space occupied by all players, it does not provide insights into spatial relationship between the two teams.

Relational space occupation of players has been investigated using *(weighted) team centroid, (weighted) stretch index*, *team spread*, *space exploration index* and *team separateness* (Figure 2). The *team centroid* is defined as the average geometrical centre of the positions of the outfield players ($n = 3$ studies).[71] The *stretch index* in longitudinal and lateral directions is defined by the average distance between each outfield player and the team centroid ($n = 4$ studies),[23,46] and gives a measure of team dispersion from the center of a team. The *weighted centroid and weighted stretch index* differ from their non-weighted counterparts by incorporating player-specific weights based on proximity to the ball, supposedly allowing a more nuanced analysis of team position and dispersion ($n = 2$ studies).[44] A derived measure is the *inter-team centroid distance* (i.e., distance between teams' centroid positions).[72] *Team spread* is defined as the Frobenius norm of all inter-player distances excluding the goalkeeper ($n = 3$ studies).[20] The *space exploration index* is the average difference between a player's current position at any given moment and their average position ($n = 1$ study).[40] *Team separateness* is calculated as a sum of distances between each player and the closest opponent [m] ($n = 1$ study).[57] There is a clear distinction in how these parameters are interpreted. Team distribution is typically defined as the extent to which a team occupies space during the course of a game, which plays a crucial role, especially in the transition between ball possession phases.[20]

When comparing the match halves, the team spread was significantly greater in the first half.[60] Spread values were higher in higher-performing teams compared with lower-performing teams.[40] Teams out of possession showed a smaller spread value when performing tackles, indicating a compact defence, whereas conceded scoring attempts involved greater spread, reflecting a less stable structure.[20] Reduced lateral and longitudinal spread contributed to effective defensive organisation.[48,49]

Stretch index values were greater against better teams and during losses than during wins and draws.[40] Furthermore, the stretch index becomes more variable throughout a match.[46] Winning teams demonstrated a greater stretch index during the attack, reflecting a broader spatial distribution.[17] The weighted stretch index decreased in the second half[44] and varied according to the score.[45] Unlike the convex hull, which considers only peripheral players, the stretch index and spread incorporate all outfield players, making them especially relevant in 11v11 contexts.

Offensive substitutions, particularly in the early stages of a losing match, shifted the team centroid towards the opponent's goal. This increased control of the attacking third and enabled more direct strategies.[73–75] The team's average field position shifted depending on the score status[45] and decreased in the second half, indicating a deeper and a more risk-averse strategy during this period.[44] The inter-team centroid distance decreased after offensive substitutions and showed smaller changes after neutral ones.[56] No significant effects were found for the space exploration index across opponent classification, outcome, status, location or time.[40] Additionally, no formation effects on team separateness were detected.[57]

Although these variables do not measure area directly, they offer insights into players' overall spatial organisation and dynamic relations with teammates.[55] For example, the inter-team centroid distance indicates team proximity and defensive strategies such as pressure or maintaining shape, while team separateness measures the distance between players and opponents, reflecting dynamic positioning.[2]

Pitch-based space conceptualisation

From a methodological perspective, dividing the pitch into distinct zones offers an alternative approach to assess player positioning. Player behaviour is mapped to pitch locations, which enables the analysis of spatial patterns based on when players enter and exit specific zones. Zubillaga et al.[68] divided the pitch into six vertical and four horizontal zones. Results show that the *total surface area* was significantly wider and shallower when the ball was in the central areas of the pitch than when the ball was in the first vertical zone extending to the height of the penalty area.[68] Cerda et al.[15] divided the pitch into thirteen zones and introduced the term *space of use* to emphasise numerical distribution within them. Their analysis revealed that most goals originated from elaborate attacks in the central penalty area involving fewer than four offensive and fewer than three defensive players.[15] Forcher et al.[49] showed that successful rest defence (despite in ball possession, certain players safe guard quick counterattacks in case of ball loss) depended on creating numerical superiority in defined defensive areas. Similarly, Praça et al.[17] introduced the concept of *near-the-ball space control*, which is a distance-based metric quantifying temporal dominance within a five-meter radius around the current ball location. In the second half, winning teams positioned more players near the ball, enabling more control of the immediate space. A problem with respect to pitch-based metrics stems from a lack of a standardised zone model. Thus, different studies use different approaches to divide the pitch into zones which makes it difficult to draw consistent conclusions across studies.

Space control in football

In contrast to pitch-based metrics, space-based metrics adapt zones to player actions. These space control-type metrics allow to assign regions of the pitch to individual players. The main idea of *space control* is that controlling specific areas of the pitch is crucial for gaining advantages, such as winning tackles and creating scoring opportunities, while effectively removing advantageous spaces from the opposing team during defense.[20]

One of the earliest space control models in football is the Voronoi approach.[26] The diagram is based on the Euclidean distance between players, thus the player closest to a certain point on the pitch is assumed to reach the location before any other player.[76] The Voronoi diagram is based on a tessellation of the pitch, where each player is assigned a distinct Voronoi cell (Figure 2).[76] A distinction is made between intra-team and inter-team space models, considering both teams separately or together.[3,77]

The concept of a Voronoi diagram is interpreted by most authors as space control ($n = 4$ studies),[56,65] but there is no clear definition. For example, Caldeira et al.[43] discuss the concept of *spatial dominance* by operationalising Voronoi diagrams, while Eliakim et al.[77] discuss the usage of space.[43,77] Memmert et al.[57,58] discuss the concept of *space control gain* (%), which reflects the change in space control between the start and end of each pass, averaged over the respective passes per trial. In the following, the concept of *space control* is delineated through the utilisation of the Voronoi diagram, wherein each player is assigned a tessellated area, and each point on the playing field is allocated to the nearest player based on the Euclidean distance between them.

Eliakim et al.[77] demonstrated higher win probability and improved performance, as measured by expected goal difference, were associated with intra-team Voronoi cell symmetry, indicating uniform player distribution, and lower maximum cut in passing networks, reflecting smoother ball circulation.[77] Comparisons between 4-2-3-1 and 3-5-2 formations showed no significant differences in space control gain,[57] yet high attention capacity teams exhibited greater gain in the final third and penalty area than low-capacity teams.[58] From a defensive perspective, numerical superiority substantially limited offensive players' space control,[49] while offensive substitutions expanded space control in the middle and attacking thirds, potentially increasing offensive success.[56] Increased space control in the attacking third correlated with passing effectiveness and match success.[65] Team formations, player roles, and ball possession further influences space control patterns.[43] A 'finishing space value' combined scoring probability from player position (distance and angle to goal) with Voronoi cell area and relative Voronoi position toward the nearest opponent, providing a reliable measure of spatial relations in finishing situations.[16] Thus, the authors used a combination of space control and a pitch-based approach. A metric that is used in combination with the Voronoi diagram is the Delaunay triangulation.[3] A Delaunay triangulation divides the convex hull into triangles (Figure 2).[78] In

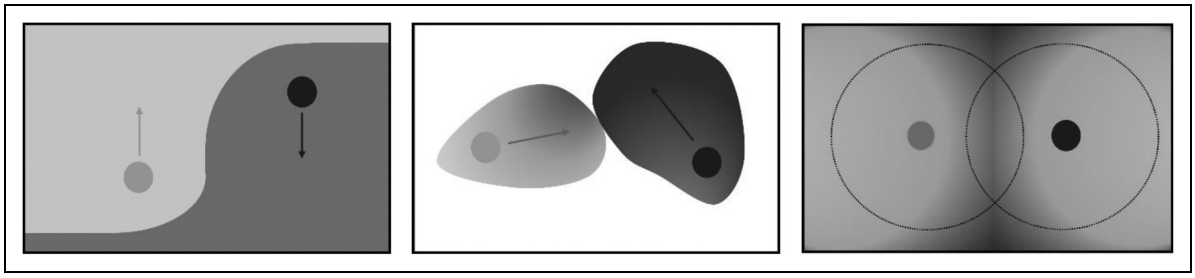

Figure 3. The images illustrate an exemplary representation of the dominant region for an individual motion model (left), a visualisation of spatial quality assessment (central), and a depiction of the corresponding overlapping area (right).

team sports, Delaunay triangulation has been employed to identify and analyse team formation structures by constructing adjacency matrices based on player positions, which can subsequently be utilised in hierarchical clustering to enable time-series analysis and quantitative comparison of team formations.[6]

In summary, evidence suggests that greater space control contributes to game success. Nevertheless, the Voronoi approach has been criticised for oversimplifying spatial control by not considering kinematic factors such as players' speed or acceleration, and thus may fail to capture the dynamic nature of football interactions.[29]

Implementation of motion models in modern football

In team sports, the evaluation of both teamwork and individual contributions is fundamentally linked to the understanding of space and spatial dynamics on the pitch.[64] A critical element in understanding spatial dynamics in football is the use of realistic motion models that capture how players navigate and control space in real time based on their individual movement profiles and predicted trajectories.[7] Thus, given a player's position on the pitch, the motion model can predict which position on the pitch can be reached by that player in a given time frame. More complex models compared to the Voronoi approach go beyond static position data by incorporating additional kinematic variables, like speed, acceleration and movement direction of the players. Naturally, individual players vary widely in their execution of movements and cannot control the entire pitch, so there are still free spaces that are not directly controlled by any player (Figure 3).

Based on this, the following section describes the various existing motion models and highlights their differences. Taki et al.[25] introduced the concept of *dominant region* based on the underlying idea that individual players dynamically control or occupy pitch zones. Under the initial definition by Taki et al.[25] a dominant region of a player is defined as a region that a player can reach before anyone else and is therefore under their control. Taki et al.[25] developed a motion model that incorporated the analysis of player positions and movement vectors, incorporating direction and speed. To this end the *minimum moving time pattern*, defined as the shortest time for a player to reach any point from their current position, along with the concept of dominant region was introduced to evaluate space management. The term '*dominant region*' is also used by Caetano et al..[42] Other authors have used different terminologies for the same concept, including 'un/coverable areas',[52] 'possible region',[53] 'pitch control',[33] 'zones of control',[29] 'space control',[78] 'sprint arrival region',[62] and 'reachable areas'.[53] The various movement models, although based on different calculations and constructs, all build on Taki et al.[25] original concept of the *dominant region*, with subsequent developments incorporating probabilistic and time-based frameworks to further refine the understanding of space in football. The term '*dominant region*' will be used further on to describe a player's ability to reach a given point from their current position in a given time.

In contrast to Voronoi diagrams, motion models allow for a more specific interpretation of space, for example by including free space which constitute a significant proportion of the entire playing field.[42] In addition, to mathematically derived deterministic models, data-driven probabilistic approaches based on machine learning and statistical modelling approaches are increasingly being used. Deterministic models rely on explicit rules and assume that future events are fully determined by present conditions, whereas probabilistic models incorporate uncertainty into their predictions by considering probability distributions over possible outcomes.[79] Caetano et al.[42] developed a motion model based on vector kinematics to analyse movement patterns and predict positions through probability matrices.[32] Fujimura and Sugihara[80] developed a two-dimensional motion model by combining physical modeling with empirical parameter fitting. Knudsen & Andersen[52] developed a methodology which determines the player's dominant region as the intersection of an ellipse defined by the movement limits and maximum displacement. Knudsen & Andersen[53] used time-step predictions at 0.5-s intervals up to 4.0 s and considered initial sprint velocities ranging from 1 to 7 m/s. Spearman et al.[66] define a model wherein the probability of a successful pass is formulated as the product of two stochastic processes: the interception probability, modelled by a logistic

distribution, and the control probability, represented by an exponential distribution.[44] The probability of a successful pass is obtained by integrating these probabilities over time and summing over all teammates except the passer.[44] Spearman et al.[31] have further developed their model to complement the ball flight time, which is determined by simulating non-linear aerodynamically influenced trajectories. Brefeld et al.[29] applied a machine learning approach using a probabilistic movement model, predicting trajectories via two-dimensional kernel density estimation at different velocities (standing, walking, jogging, running and sprinting). Gu et al.[51] developed a deep generative model that integrates tracking and event data to analyse the spatial efficiency, showing that its outcome closely correlates with actual team performance and match success.

Taki and Hasegawa[67] found that effective player's movement extends a team's dominant region and high quality passing occurring within a teammate's dominant region.[67] Knudsen and Andersen[52] showed that at certain passing speeds, an opponent's defence is well coordinated with no gaps, but if the passing speed increased, gaps may appear that the defence cannot close.[52] Knudsen and Andersen[53] showed spatiotemporal asymmetries in a team playing a 4-1-4-1 formation with medio-lateral shifts (0.5–3%) in the center of dominant regions (4–29 cm) towards the favoured side, influenced by changes in direction, physiological asymmetries, positions or tactical considerations. Knudsen et al.[54] further demonstrated that the space that football players can occupy changes significantly based on initial speed and time, with the dominant regions being unique in shape and size based on the player's morphology. Spearman et al.[53] evaluated the quality of off-ball positioning, preceding shots, that could lead to goals and the probability of scoring from a position, assuming the offensive team successfully controls the ball at that position.

Beyond these concepts, assigning discrete dominant regions to individual players in football is challenging, as roles and responsibilities often overlap. This overlap makes it difficult to clearly define a unique area for each player, raising questions about the clarity and precision of region-based assessments. Allegre and Vuillemot[30] defined the term *disputed areas* as those areas on the pitch that can be reached by two or more players at the same time (Figure 3). The approach is based on two occupation models: the *temporal distance model* by Rolland et al.,[81] which takes orientation and speed of the players into account, and the model by Stein et al.,[82] which is used to remove areas close to the players. This model has been chosen to automatically detect region-based 'faulty movement behaviour', offering valuable insight into assists, particularly the final pass before a goal, and was informed by expert feedback regarding the relevance and presentation of these areas.[82]

In summary, dominant regions are a dynamic concept that reflects individual skills and movement, whereas the classic Voronoi diagram offers a static, position-based view. While regions match when players are stationary, higher speeds cause elliptical divergence, exposing limitations in capturing dynamic football interactions.[53,67] Probabilistic methods employ various algorithms and visualisations to illustrate the underlying calculations and apply these principles to tactical spatial analysis in football. This integrated approach enhances understanding of player movement, strategies and pitch interactions by drawing on dynamic models, field weighting and advanced, data-driven evaluation.

Evaluation of motion models

A major problem with motion models used to quantify space in football is the frequent lack of explicit validation of these models. One way of validating probabilistic motion models is to categorise passes as either successful and unsuccessful.[6,51] A prediction is confirmed when the ball is received by the predicted player in the predicted area, and it fails if another player receives the ball. Narizuka et al.[6] validated the Fujimura & Sugihara[80] motion model by comparing its predicted arrival regions with empirical position data under sprinting conditions. Caetano et al.[42] demonstrated that model accuracy in predicting dominant regions decreases over time, correctly predicting 96.6%, 88.6% and 72.3% for one, two and three seconds.[42] Probabilistic models have been compared to other methods to show improved accuracy in identifying dominant regions.[29] Another approach validated models by correlating off-ball scoring opportunities with future scoring rates, indicating reduced stochasticity.[31] Expert assessment based on relevance, visual design, and analytical context was used to validate these approaches.[30,82] In summary, there are multiple approaches to evaluating motion models. However, validating these models is essential yet often neglected in studies. The nature of the research findings, together with the lack of appropriate model validation which is prevalent across the reviewed studies, highlights the critical need for a more rigorous evaluation of their actual contribution to understanding football performance in practical applications.

Strength and limitations

The selection criteria and data collection cut-off was set at August 2025, so later studies are missed from this review. A meta-analysis was not possible due to heterogeneity in measures, reporting standards, seasonal contexts, match conditions, and use of different data types such as GPS and (semi-) automated video tracking at different performance levels. Most studies did not mention or could not verify, the validity or reliability of the outcome measures, which must be considered critically in the analysis. Also, most studies in this field of research do not follow the standard PICO model, that makes a systematic comparison of

methodical approaches and outcomes particularly difficult. However, this review provides transparent reporting and a thorough quality assessment, including risk of bias, which enables critical evaluation and comparison of the evidence.

Conclusions

This review synthesised 46 studies on spatial analysis in football, focusing on the conceptualisation, construction and application of 'space' in competitive football. The review revealed a variety of methodologies, which can be broadly categorised as deterministic or probabilistic. Deterministic models use fixed calculations, such as Euclidean distances or pitch zones, whereas probabilistic models incorporate player kinematics and historical data to estimate dynamic dominant regions. Notably, identical spatial values can arise from structurally different team formations, and inconsistent terminology complicates cross-study comparisons. Many studies exhibited methodological weaknesses, particularly in terms of lacking consistency of terminology, transparency of analysis, and validation of applied models. These limitations highlight the restricted practical relevance of current spatial metrics in football analysis and emphasise the need for future research to realise their substantial potential in practical applications. Nevertheless, spatial metrics have a practical application in contemporary football, supporting performance monitoring, tactical adjustments, and tailored training. This review clarifies these methodological intricacies and their empirical significance. Future research should specify the context in which each spatial variable is most meaningful, reduce complexity and identify the variables that best capture tactical nuances. This would enhance the precision and usefulness of spatial analysis in football.

ORCID iDs

Lisa Krampe https://orcid.org/0009-0001-9028-8380
Dominik Raabe https://orcid.org/0000-0001-7264-4575

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